Dar es Salaam Institute of Technology (DIT)

Industrial Practical Training (IPT) Report

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Declaration

I, Emmanuel Katayi Luanga, do hereby declare that this report is my own original work carried out during the Industrial Practical Training at Dar es Salaam Institute of Technology and has not been submitted elsewhere for any academic award.

Signature: ____________________

Date: ____________________

Abbreviations

Abbreviation	Meaning
DIT	Dar es Salaam Institute of Technology
IPT	Industrial Practical Training
CPU	Central Processing Unit
LAN	Local Area Network
RBAC	Role-Based Access Control
SMTP	Simple Mail Transfer Protocol
ICT	Information and Communication Technology
SMS	School Management System
UI	User Interface
DB	Database
2FA	Two-Factor Authentication

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CHAPTER 1: INTRODUCTION

1.1 Background of the Organization

Dar es Salaam Institute of Technology (DIT) is a leading higher learning institution in Tanzania, specializing in engineering and technology disciplines. Its Information and Communication Technology (ICT) environment is well-equipped with modern computer labs, reliable Local Area Networks (LAN), servers for internal hosting, and a rich repository of educational software.

DIT encourages practical learning, and its ICT unit provides a strong technical infrastructure to support students' real-world applications. During my Industrial Practical Training, I experienced a highly organized environment that fosters innovation, learning, and technical skill development.

1.2 Organization Structure

The hierarchy at DIT is designed to ensure smooth supervision and accountability:

- **Principal**: Overall leadership and policy direction.
- **Heads of Departments (HoDs)**: Oversee departmental functions and programs.
- **ICT Unit**: Responsible for ICT infrastructure, software deployment, and user support.
- **Technicians**: Handle day-to-day maintenance of hardware and network systems.
- **Student Trainees**: Undertake practical assignments, reporting to supervisors for guidance.

This structure promotes a balance between administrative oversight and hands-on learning.

1.3 Vision, Mission, and Objectives

1.3.1 Vision

To be a center of excellence in technology and innovation, producing highly skilled ICT professionals equipped to address real-world challenges.

1.3.2 Mission

To provide practical, high-quality training in ICT through modern facilities, mentorship, and industry-oriented projects, enabling students to develop professional competence and problem-solving skills.

1.3.3 Objectives

- Equip students with hands-on ICT skills.
- Foster understanding of enterprise system development.
- Promote innovation and technical problem-solving.
- Enhance teamwork and professional conduct.

CHAPTER 2: ACTIVITIES PERFORMED

2.1 Overview of the Undertaking

The main assignment during my 10-week IPT was to **develop a comprehensive School Management System (SMS)** for DIT. The system was designed with multiple subsystems and role-based access, aimed at enhancing administrative efficiency in school management.

The SMS project integrated key technical skills: **PHP development, MySQL database management, RBAC implementation, session handling, subsystem management, and LAN deployment**.

2.2 Activities Performed

2.2.1 Database Design and Implementation

- Designed MySQL schemas for multiple subsystems including **Library, Student Information, Inventory, Transport, and ICT**.
- Established relationships and constraints to ensure data integrity.
- Created backup and migration plans for smooth deployment.

2.2.2 User Management Module Development

- Implemented features to **create, edit, and manage user accounts**.
- Role assignment functionality: **Super Admin, Subsystem Admin, Normal User**.
- Ensured secure login, logout, and session management.
- Enabled password resets and account activation/deactivation.

Table 2.1: User Roles and Subsystem Access

Role	Subsystem Access	Permissions

Super Admin	All	Full system control
Subsystem Admin	Assigned subsystem	CRUD operations
Normal User	Assigned subsystem	Limited access

2.2.3 Role-Based Access Control (RBAC)

- Implemented requireRoleByName() function to restrict access.
- Customized dashboards per role.
- Handled unauthorized access by redirecting users to appropriate pages.

2.2.4 Subsystem Interfaces Development

- Developed subsystem-specific dashboards with personalized greetings.
- Each subsystem admin could view only their assigned modules.
- Integrated Library, Student Info, Inventory, Transport, and ICT subsystems.

2.2.5 Super Admin Dashboard

- Full visibility of all subsystems and roles.
- Monitored activities via audit logs.

- Managed global settings including maintenance mode and theme configurations.

2.2.6 Advanced System Settings Module

- **General Settings**: System name, logo, timezone, language, default role, system status.
- **Theme/UI Settings**: Light/Dark mode, sidebar style, layout options.
- **Security Settings**: Password policy, session timeout, login attempt limit, 2FA toggle.
- **Email/SMS Settings**: SMTP configuration, sender details, SMS gateway.
- **Backup/Data Management**: Manual and scheduled backups, restore operations.
- **Audit Logs**: Log retention periods, activity monitoring.

Table 2.2: System Settings Overview

Setting Type	Features
General	Name, logo, timezone, language, status
Theme/UI	Color schemes, layout style, mode
Security	Password policy, 2FA, session timeout
Email/SMS	SMTP, sender, gateway

Backup	Manual/auto backups, restore
Audit Logs	Retention, monitoring

2.2.7 Deployment & Testing

- Hosted SMS on local DIT LAN.
- Ensured accessibility from both PCs and mobile devices.
- Conducted unit testing, debugged PHP and SQL errors.
- Verified RBAC enforcement and subsystem isolation.

2.2.8 Network Management

- Configured local hosting and database connections.
- Managed backups and ensured system resilience.

Figure 2.1: Example of SMS Dashboard
Figure 2.2: User Management Interface
Figure 2.3: System Settings Page

2.3 General Observations

- DIT has well-organized ICT infrastructure, enabling smooth development and deployment.
- Supervision was consistent; both institute and industrial supervisors provided guidance and constructive feedback.
- Teamwork, even among student trainees, enhanced problem-solving and learning.
- Daily workflow emphasized documentation, testing, and system security.

2.4 Challenges

- **Technical**: Session management conflicts, SQL query errors, and RBAC bugs during subsystem integration.
- **Operational**: Limited access to certain network resources; occasional hardware unavailability.

2.5 How Challenges Were Solved

- Debugged PHP code and refined SQL queries iteratively.
- Consulted supervisors for best practices in session handling and RBAC implementation.
- Scheduled tasks during off-peak hours to access limited network resources.
- Created detailed logs for tracking system issues.

CHAPTER 3: CONCLUSION & RECOMMENDATIONS

3.1 Conclusion

The 10-week Industrial Practical Training at DIT provided an immersive experience in real-world ICT operations and enterprise system development. Developing the **School Management System** strengthened my skills in PHP programming, database management, RBAC, and system deployment.

I gained practical exposure to team collaboration, problem-solving under supervision, and handling technical and operational challenges. Overall, the experience has significantly enhanced my professional competence as an IT student.

3.2 Recommendations

For DIT:

- Expand access to advanced development tools and frameworks.
- Increase hardware resources in ICT labs for simultaneous student projects.

For Future Students:

- Focus on documenting processes thoroughly.
- Engage actively with supervisors to understand real-world constraints.
- Develop test plans early to identify issues during implementation.

References

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3. Tanenbaum, A., & Wetherall, D. (2011). *Computer Networks* (5th Edition). Pearson.

4. Microsoft Docs. (2024). *Role-Based Access Control (RBAC) Overview*. https://docs.microsoft.com

Appendices

- **Appendix A**: Logbook Summary
- **Appendix B**: Screenshots of SMS Dashboards and Subsystems
- **Appendix C**: Supervisor Signatures